Real-time 2D Direction Steered Beamforming with a minimal 3 microphone array on a Resource-constrained FPGA

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Abstract

Directional hearing and assistive-listening devices benefit from knowing where a speech source is, but existing FPGA sound-localization systems typically use large microphone arrays, offload part of the pipeline to a hard processor or DSP, or report only coarse direction. This preliminary report surveys FPGA time-delay-of-arrival (TDOA) localization and beamforming designs, direction-of-arrival (DOA) algorithms, and hardware constraints in hearing-aid applications. We find that no reviewed design combines a minimal three-microphone array with fully embedded two-dimensional direction estimation on a small FPGA. We therefore propose an implementation and resource-characterization study: a processor-free RTL pipeline on a Xilinx Spartan-7 XC7S50 comprising I2S capture from three microphones, time-domain GCC-PHAT-style TDOA estimation for each microphone pair, azimuth estimation from the pairwise delays, and a DOA-steered delay-and-sum beamformer with real-time audio output and a bearing display. We will report LUT, DSP, and BRAM utilization against surveyed devices, together with angular accuracy. Stretch goals include a software baseline, a frequency-domain variant, adaptive nulling, and continuous tracking.

1. Introduction

We began our discussions with sound and acoustic signal processing applications, as there seemed to be a lot of scope in this area that was doable for us to cover. Eventually we narrowed it down to signal processing applications for hearing aids [24]. Of these, the most interesting was the ability to zoom into a particular direction, or in other words, find the direction of a speech source. Inspired by this, we decided to focus on minimal implementations of this concept on limited FPGAs, similar to the hardware constraints we are operating under. We attacked the related literature from different angles, separating the overall literature review into multiple pieces.

2. Foundational Sound Source Localisation (SSL) Algorithms & Their FPGA Implementations

The method of time-delay-based sound localization started from the Generalized Cross-Correlation (GCC) method of Knapp and Carter [20], whose PHAT weighting is still considered the standard way to estimate delays reliably in noisy, echoey rooms. An early FPGA system to use this idea was built by Nguyen et al. [27] in 2003: it was a two-microphone localizer that ran entirely on one chip and worked down to 10 dB SNR, with several such modules fitting on a single device. Over the next decade, three- and four-microphone FPGA designs mostly focused on making the hardware smaller and cheaper rather than more accurate. For example, a cross-correlation system with a separate azimuth stage [17], a soft-processor-based design using the MCCC algorithm [39], and a compact SOPC-based localizer [23], all reporting errors under about 3°. A related line of work looked at beamforming instead of localization, such as a fixed-null, two-microphone hearing-aid beamformer [34], which steers a null toward noise but does not track a moving source. The 2018 survey by Da Silva et al. [6] shows that FPGA microphone-array research after this point mostly scaled up to tens of microphones for delay-and-sum beamforming, rather than pushing small arrays further. In a recent FPGA-based Time-Delay-of-Arrival (TDOA) result we found, Jamaludin et al. [16], reports a mean angular error of $\pm 2.29°$ (97.46 % accuracy) on a DE1-SoC, though this was tested with a single tone rather than real speech, so it is not directly comparable to speech-based systems. Outside FPGA work, Kovalyov et al. [21] built the closest match to our goal, it is a three-microphone system that estimates 2D direction and tracks a moving source ,but on a smartphone rather than dedicated hardware, and Grondin et al. [12] showed that a simplified cross-correlation method can cut computation cost further on small microphone arrays, supporting our choice to avoid the frequency-domain approach (see Section 3).

Table 1. FPGA-based TDOA/SSL designs surveyed, in chronological order.

Work	Year	Algorithm	Mics	Device	Reported accuracy / result
Nguyen et al. [27]	2003	Modified-PHAT cross-correlation	2 (×5 parallel)	Virtex-II class	10 dB SNR floor; 77–108 mW/mic
Sound loc. system [17]	2008	Cross-corr. + energy + azimuth	3	Not reported	Real-time; 3°
MCCC DOA [39]	2013	MCCC (soft-core)	4	Cyclone II + Nios II	$< 3°$ deviation
Compact SOPC device [23]	2016	TDOA (SOPC)	Not reported	Cyclone III	$< 3°$ deviation
Fixed-null beamformer [34]	2017	Fixed-null beamforming (no DOA tracking)	2	Artix-7 (xc7a100t)	Resource utilization only (Vivado)
Da Silva et al. survey [6]	2018	Survey (multiple)	8–300 (surveyed)	Multiple	Survey; see Table 2–5 of source
Kovalyov et al. [21]	2023	mCPSP TDOA + NNS + tracking	3	Smartphone SoC (not FPGA)	Sub-lattice RMSE; qualitative tracking
Jamaludin et al. [16]	2024	TDOA	Not reported	DE1-SoC	$\pm 2.29°$ (97.46%)
This work	2026	Time-domain GCC-PHAT + 2D DOA steered beamforming	**3**	**Spartan-7 (XC7S50)**	*In progress*

3. FPGA Architecture and Resource-Efficient Implementation

Beyond the choice of localization algorithm, the surveyed FPGA-based systems diverge sharply in how much of the acoustic pipeline is delegated to a hard processor core versus implemented entirely in reconfigurable fabric. Sledevič and Laptik [35] demonstrate a representative SoC-class partitioning: their Xilinx Zynq-7020 design computes the six pairwise cross-correlations for a four-microphone GCC-TDOA array entirely in the programmable logic, while the embedded ARM Cortex-A9 core resolves the resulting delays into a 3-D source position, completing one full estimate in 28 ms. A similar split is used by Kowalczyk et al. [22], whose SoC FPGA performs demodulation and filtering on-chip before handing off SRP-PHAT localization and MVDR-based speech enhancement to the hard core, and by Azcarreta [2], whose 48-microphone Filter-and-Sum array uses its hard core only for network communication rather than for any signal-processing task. Takagi et al. [36] extend this pattern to a distributed setting, where each 16-element SoC-FPGA node runs a voice-activity detector and the MUSIC algorithm on-chip and reserves the hard core for inter-node communication. In contrast, an equally active line of work avoids a hard processor altogether. Petrica [29, 30] implements an 8- and later 16-element Delay-and-Sum beamformer entirely on bare fabric (Microsemi IGLOO2, then Xilinx Spartan-6), reporting 21.8 mW and 61.71 mW respectively with no host-side computation, and da Silva et al. [6] push this further to a 52-element array at 34 mW on a Flash-based SmartFusion2 device. Bare-fabric TDOA localization persisted at small scale as well: Biswas et al. [3] implement a coherence-based two-microphone TDOA locator entirely on a Xilinx Spartan-6, with no SoC or host involvement, as late as 2017. Thus, it was contemporaneous with, rather than preceding, the SoC-based designs above. The one three-microphone system in this body of work, the countersniper array of L'edeczi et al. [25], is itself bare-fabric (Xilinx XC2S100, using only a PicoBlaze soft-core for Bluetooth communication), but performs discrete shockwave/muzzle-blast event classification rather than continuous angle estimation. Taken together, this indicates that the SoC-versus-bare-fabric choice in the reviewed literature tracks array size and task complexity rather than chronology, and that no design – SoC or bare-fabric – combines a minimal three-microphone array and continuous, fully-embedded 2D DOA tracking on an extremely resource-constrained FPGA.

The choice between time-domain and frequency-domain beamforming fundamentally dictates the hardware footprint of FPGA sound locators, as detailed by da Silva et al. [7]. Frequency-domain processing forces the FPGA to execute multiple Multiply-Accumulate (MACC) operations to apply phase shift corrections, which mandates complex data representations using high bit widths or floating-point numbers [7]. To avoid the significant time and hardware resources demanded by the discrete Fourier transformation, time-domain Delay-and-Sum beamforming is widely preferred, as it eliminates these heavy MACC operations by simply using internal memory blocks to delay and synchronize incoming audio samples [6, 7]. This domain shift drastically alters the primary hardware bottlenecks. For instance, a high-performance 52-microphone architecture deployed on a Zynq 7020 FPGA is constrained primarily by logic, consuming 79.5% of the available Look-Up Tables (LUTs) and 41.8% of the DSP48 blocks. However, a power-efficient architecture on the same Zynq 7020 drastically reduces this logic footprint to just 7.6% for LUTs and 2.7% for DSP48s, shifting the primary constraint to internal memory by utilizing 18.9% of the BRAM18K resources [7]. When targeting a low-power Microsemi Smart-

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Fusion2 M2S025 FPGA, this power-efficient design consumes exactly 5,547 4LUTs, 6 MACC (DSP) units, 216 RAM64x18 blocks, and 28 RAM1K18 blocks [7]. Overall, these established tradeoffs highlight how time-domain memory utilization replaces MACC logic complexity, providing the benchmark for us to choose time-domain processing on FPGAs that are primarily constrained in LUT count and other resource baselines.

4. Hardware constraints in assistive listening

Microphone-array design for hearing-aid and assistive-listening applications involves a trade-off between spatial selectivity, physical size, and computational complexity. Two-microphone directional systems have long been attractive because of their compact form factor; FPGA-based work has demonstrated adaptive beamforming with only two omnidirectional microphones while remaining suitable for real-time implementation [33, 34]. However, increasing the number of microphones can provide additional spatial degrees of freedom. Powers and Hamacher demonstrated a three-microphone behind-the-ear directional instrument, reporting an improvement in directivity over conventional two-microphone processing, while also identifying practical limitations arising from microphone mismatch, low-frequency attenuation, microphone noise, and the small spacing available in a hearing-aid form factor [31,32]. More generally, microphone-array design for hearing aids is constrained by device dimensions, computational resources, and subject-dependent acoustic effects, which limit the practicality of simply scaling the array size [1]. At the other extreme, Inoue *et al.* [15] demonstrated a 48-microphone FPGA-based hat-type hearing-aid system, illustrating the directivity and steering flexibility achievable with a large aperture. The hat-based form factor, however, differs substantially from the compact physical constraints of ear-level hearing devices. Recent work has therefore increasingly focused on adaptive spatial processing rather than microphone count alone: multibeam hearing-aid systems can track multiple conversational partners in real time [9], while virtual-microphone methods seek additional spatial degrees of freedom from only a small number of physical microphones [8]. In this context, three microphones provide an intermediate design point: unlike conventional two-microphone beamformers, a non-collinear three-microphone geometry provides sufficient TDOA measurements for two-dimensional direction estimation, while avoiding the substantial physical and computational overhead of large arrays.

5. Comparison of DOA Algorithms

TDOA/DOA estimation is approached on accuracy, robustness, and computational grounds. Brutti *et al.* [4] compared GCC-PHAT against Adaptive Eigenvalue Decomposition (AED) and found AED more accurate under clean, non-interfering conditions. However, due to the iterative nature of its eigenvector estimation, AED was found to be considerably more computationally demanding than GCC-PHAT. This iterative loop makes it a poor fit for a fixed-resource FPGA pipeline, and GCC-PHAT matched or exceeded AED's accuracy on SNR below $15\,\mathrm{dB}$. Nguyen and Pham [28] compared classical cross-correlation (CC), the smoothed coherence transform (SCOT), GCC-PHAT, and SRP-PHAT for multi-source detection-localization. All three GCC-family methods were computationally cheaper than SRP-PHAT, though SRP-PHAT achieved better localisation precision at the cost of an exhaustive grid search. Such a method is not viable with a limited number of DSP slices. Kadhim and Khalifeh [18] benchmarked MUSIC, SRP-PHAT, WAVES, CSSM, FRIDA, and TOPS for hearing-aid type embedded systems, finding MUSIC and SRP-PHAT fastest ($0.030\,\mathrm{s}$ and $0.046\,\mathrm{s}$ respectively) while CSSM, WAVES, TOPS, and FRIDA required $0.47\,\mathrm{s}$ to $1.17\,\mathrm{s}$. However, methods involving subspace-decomposition and sparse-reconstruction must be discarded for real-time FPGA pipelines, since they reflect eigendecomposition or search costs that are extremely expensive on a 120-DSP-slice device. Finally, as recently as 2025, Choi *et al.* [5] proposed a filter-bank-weighted kernel density estimation as a noise-robustness extension of GCC-PHAT. This shows that GCC-PHAT is still actively used in research, and is not a superseded method.

One alternative we encountered deserves explicit acknowledgment: Ying *et al.* [40] propose a weighted minimum mean square error (WMMSE) estimator that avoids grid search entirely, reporting robustness close to SRP-PHAT at roughly five times the speed of GCC-PHAT and thirty times the speed of SRP-PHAT in software. WMMSE has not, to our knowledge, been validated in an FPGA or other embedded RTL implementation, so adopting GCC-PHAT (the method used by every hardware implementation reviewed in Section 2) carries lower implementation risk. However, we note WMMSE as a candidate for future work or a stretch goal due to its promising potential.

6. Minimal-Array Systems

Real-time 2D DOA tracking using a minimal three-microphone array is a well published direction, especially framed for hearing-aid style directional listening. Gu and Yang [13] demonstrated a three-microphone crosspower spectrum-phase system achieving $4.02°$ azimuth and $2.18°$ elevation accuracy. Independently from them, a long research program at the University of Texas at Dallas also progressed over 7 years. They began with an SVD-based non-uniform three-microphone smartphone system [10] through GCC and RSVD-based L-shaped-array implementations. This was developed targeting hearing-aid

Table 2. SoC-class vs. bare-fabric task partitioning in surveyed FPGA microphone-array designs.

Ref.	Year	Mics	On-FPGA fabric	On hard core / host	Power
[35]	2017	4	GCC cross-correlation (6 pairs)	3D position (28 ms)	N/S
[22]	2016	7	Demod. + filtering	SRP-PHAT + MVDR	N/S
[2]	2016	48	Filter-and-Sum beamform.	Network comms. only	N/S
[36]	2010	16	VAD + MUSIC + D&S	Inter-node comms.	N/S
[30]	2015	8	Full pipeline (bare, IGLOO2)	—	21.8 mW
[29]	2016	16	Full pipeline (bare, Spartan-6)	—	61.71 mW
[6]	2018	52	Full pipeline (bare, SmartFusion2)	—	34 mW
[3]	2017	2	Full TDOA pipeline (bare, Spartan-6)	—	N/S
[25]	2005	3	Shockwave/muzzle-blast detect (bare)	PicoBlaze soft-core (comms. only)	N/S

users [37, 38]. Recently, their 2023 system [21] adds a learned time-difference-of-arrival (TDOA)-to-DOA mapping, multi-source clustering, and demonstrated tracking of a moving source. This covers most of the features that we plan to implement, plus some other additional ones. Finally, the most recent related work that we identified: Lopez-Olvera *et al.* [26] reported a four-microphone circular-array GCC-PHAT-and-delay-and-sum beamforming system explicitly designed for resource-constrained embedded platforms. They achieved sub-1° accuracy at 3.47 ms per-frame processing time. They published in August 2026, just a few weeks before this report. We took this as an indication that the field is still highly active.

We also analysed two broader surveys: Da Silva *et al.*'s [6] review of FPGA-based acoustic beamforming architectures has already been covered earlier. Other than that, there is Grinstein *et al.*'s [11] tutorial review of over 200 steered-response-power (SRP) papers. The latter noted that device-specific SRP implementations spanning CPUs, GPUs, and FPGAs achieve comparable performance. In conclusion, hardware realisations of these algorithms do exist, but they mostly target large mic arrays and SoC class platforms or general purpose DSP modules (like the XMOS chip used by Lopez-Olvera *et al.*), instead of working with bare, processor-less FPGA fabric.

Separately, Hu *et al.* [14] independently confirm that minimal three-pair microphone configurations yield relatively large estimation error relative to larger arrays, matching earlier quantitative findings by Klein and Vo [19], who showed three-microphone TDOA accuracy dropping to approximately 55% at $-10\,\mathrm{dB}$ SNR versus 99% for a ten-microphone array.

Taken together, we present this project as an implementation and resource characterisation study: realising this established three-microphone GCC-PHAT DOA steered beamforming pipeline entirely in RTL on a bare, SoC-less, processor-free FPGA, and reporting the resulting resource utilisation and accuracy trade-offs against the DSP and smartphone based implementations that otherwise dominate this space.

7. Motivation for a Bare FPGA Implementation

As the main contribution of our work, the value of a bare, SoC-less FPGA realization over the DSP and smartphone based implementations needs defending.

First, RTL implementations are the established precedent for migrating an assistive-listening algorithm toward a custom low-power ASIC. This was the path taken by Samtani *et al.* [34], whose FPGA hearing-aid beamformer was followed by an ASIC-optimized version of the same design the same year [33]. On the other hand, a software implementation on a commercial DSP module has no equivalent path.

Secondly, a fixed hardware pipeline offers cycle level deterministic latency and substantially lower power per operation than a general-purpose core running the same algorithm in software. These are both relevant parameters to a battery-powered wearable device. Finally, the parameter we wish to measure is that the actual size of this trade-off for the specific GCC-PHAT-and-delay-and-sum pipeline under study has not been measured anywhere in the reviewed literature.

8. Problem Statement

With the above analysis, we finally conclude the statement of our project.

8.1. Core Goals

- 3-microphone triangular array with I2S audio capture via a custom Register-Transfer Level (RTL) receiver
- Time-domain Phase Transform Generalized Cross-Correlation (GCC-PHAT)-style Time Difference of Arrival (TDOA) calculation for each microphone pair
- 2D Direction of Arrival (azimuth) estimation derived from the three pairwise TDOAs
- Live DOA-steered Delay-and-Sum Beamformer (DSB), outputting real-time audio through the onboard audio jack

- HDMI and/or LED-arc bearing indicator for visualisation
- Comprehensive resource utilization report (LUT, DSP, and BRAM consumption mapped against the physical limits of the Xilinx Spartan-7 XC7S50 FPGA), positioned against surveyed device benchmarks.

8.1.1 Novelty Claim

We characterise a 2D-DOA-tracked acoustic beamformer using a 3-microphone array on a severely resource-constrained device class currently absent from surveyed literature.

8.1.2 Next Steps

1. **Three-Microphone I2S Audio Capture**
   - **Study & Design:** Understand I2S protocol, microphone interfacing, clock generation, and RTL-based serial data reception.
   - **Implementation:** Design and implement the I2S receiver and microphone interface in Verilog/RTL, including synchronization and buffering.
   - **Testing & Validation:** Test captured samples against known audio/data patterns and understand reliable simultaneous acquisition from all three microphones.
2. **GCC-PHAT / TDOA Estimation**
   - **Study & Design:** Must look into cross-correlation, GCC-PHAT, phase transform, and methods for estimating TDOA between microphone pairs.
   - **Implementation:** Develop a hardware-efficient time-domain TDOA algorithm for the three microphone pairs.
   - **Testing & Validation:** Compare FPGA-derived TDOAs against software calculations and evaluate accuracy under different source positions and noise conditions.
3. **2D Direction-of-Arrival Estimation**
   - **Study & Design:** Build a system between the three pairwise TDOAs and the source azimuth for the triangular microphone geometry.
   - **Implementation:** Implement the azimuth estimation algorithm using the measured TDOAs.
   - **Testing & Validation:** Test the estimated bearing at different known source angles and calculate angular estimation error.
4. **Live DOA-Steered Delay-and-Sum Beamformer**
   - **Study & Design:** Study delay-and-sum beamforming, steering delays, interpolation/quantization requirements, and FPGA implementation constraints.
   - **Implementation:** Build the real-time beamformer and integrate it with the estimated DOA.
   - **Testing & Validation:** Evaluate the beamformer's ability to enhance the desired direction and suppress signals arriving from other directions.
5. **HDMI / LED-Arc Bearing Visualization**
   - **Study & Design:** Determine the required display/LED interface and design the mapping between estimated azimuth and visual bearing.
   - **Implementation:** Implement the HDMI /or LED-arc visualization system and integrate it with the real-time DOA output.
   - **Testing & Validation:** Verify that the displayed bearing follows the actual source position with minimal latency.
6. **FPGA Resource Utilization and Benchmarking**
   - **Study & Design:** Study the resource requirements of the proposed architecture.
   - **Implementation:** Synthesize and integrate the individual modules and obtain their resource utilization.
   - **Testing & Validation:** Generate a comprehensive utilization report and compare the obtained resource consumption with the surveyed device benchmarks, while identifying optimization opportunities.

8.2. Stretch Goals/Future Work

The following extensions will be implemented sequentially (arranged in order of novelty/feasibility) if the core goals are finished ahead of time. Otherwise, we will consider this for future work.

1. **Software Baseline Comparison:** Run the identical GCC-PHAT-and-delay-and-sum pipeline on a general-purpose low-power processor (such as a Raspberry Pi) and report latency/resource/power side-by-side with the FPGA RTL version, on the same 3-mic array.
2. **Frequency-Domain Variant:** FFT-based GCC-PHAT variant built in parallel with the time-domain pipeline to facilitate a direct empirical comparison of resource usage versus directional accuracy.

3. **Formal Accuracy Characterization:** Systematic experimental sweep quantifying angle estimation error against ground-truth angle, microphone array geometry/spacing, and Signal-to-Noise Ratio (SNR).

4. **Adaptive Nulling:** Basic adaptive null-steering oriented toward a secondary non-target direction, advancing beyond fixed-null baselines toward dynamic spatial filtering.

5. **Continuous Target Tracking:** Implementation of a tracking and smoothing filter (e.g., Kalman/Alpha-Beta) to follow moving speakers, replacing snapshot-only DOA estimates with continuous trajectory tracking.

6. **Directional Stereo Panning:** Implement stereo audio to acoustically show the direction of an audio signal to the user (as done in hearing aids that preserve binaural cues) instead of collapsing to mono as other literature does.

7. **WMMSE Implementation:** RTL implementation of the WMMSE as proposed by Ying *et al.* [40] and compare with our GCC-PHAT implementation.

References

[1] D. Ayllón, R. Gil-Pita, and M. Rosa-Zurera. Design of microphone arrays for hearing aids optimized to unknown subjects. *Signal Process.*, 93(11):3239–3250, Nov. 2013. 3

[2] Juan Azcarreta Ortiz. Pyramic array: An fpga based platform for many-channel audio acquisition. Master's thesis, École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland, 2016. 2, 4

[3] Tanmay Biswas, Sudhindu Bikash Mandal, Debasri Saha, and Amlan Chakrabarti. Coherence based dual microphone speech enhancement technique using fpga. *Microprocessors and Microsystems*, 55:111–118, 2017. 2, 4

[4] A. Brutti, M. Omologo, and P. Svaizer. Comparison between different sound source localization techniques based on a real data collection. In *Proc. Hands-Free Speech Commun. Microphone Arrays (HSCMA)*, pages 69–72, 2008. 3

[5] J. Choi, J. Kim, and S. Im. FB-WKDE: A robust sound source localization approach for TDOA estimation in noisy environments. *IEEE Access*, 13:177745–177759, 2025. 3

[6] Bruno Da Silva, An Braeken, and Abdellah Touhafi. FPGA-based architectures for acoustic beamforming with microphone arrays: Trends, challenges and research opportunities. *Computers*, 7(3):41, 2018. 1, 2, 4

[7] Bruno da Silva, Laurent Segers, An Braeken, Kris Steenhaut, and Abdellah Touhafi. Design exploration and performance strategies towards power-efficient fpga-based architectures for sound source localization. *Journal of Sensors*, 2019:1–27, 2019. 2, 3

[8] Mojtaba Farmani, Svend Feldt, and Jesper Jensen. Beamforming using virtual microphones for hearing aid applications. In *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 15552–15556, 2026. 3

[9] P. Folkeard, N. S. Jensen, H. Kamkar Parsi, S. Bilert, and S. Scollie. Hearing at the mall: Multibeam processing technology improves hearing group conversations in a real-world food court environment. *Am. J. Audiol.*, 33(3):782–792, Sept. 2024. 3

[10] A. Ganguly, C. Reddy, Y. Hao, and I. Panahi. Improving sound localization for hearing aid devices using smartphone assisted technology. In *Proc. 2016 IEEE Int. Workshop Signal Process. Syst. (SiPS)*, pages 165–170, 2016. 3

[11] E. Grinstein, E. Tengan, B. Çakmak, T. Dietzen, L. Nunes, T. van Waterschoot, M. Brookes, and P. A. Naylor. Steered response power for sound source localization: A tutorial review. *EURASIP J. Audio, Speech, Music Process.*, 2024(1):59, 2024. 4

[12] François Grondin et al. Fast cross-correlation for tdoa estimation on small aperture microphone arrays. In *arXiv preprint*, 2022. 1

[13] H.-Y. Gu and S.-S. Yang. A sound-source localization system using three-microphone array and crosspower spectrum phase. In *Proc. 2012 Int. Conf. Machine Learning and Cybernetics*, pages 1732–1736, Xi'an, China, 2012. 3

[14] X. Hu, J. Zhang, H. Ma, J. Shi, and Y. Du. Robust real-time DOA estimation for outdoor vehicle acoustic sources using dynamic-pruning GCC-PHAT and adaptive forgetting factor OPAST-MUSIC. *Sensors*, 26(13):4281, 2026. 4

[15] T. Inoue, Y. Ikeda, and Y. Oikawa. Hat-type hearing aid system with flexible sound directivity pattern. *Acoust. Sci. Technol.*, 39(1):22–29, Jan. 2018. 3

[16] Norish Camelia Jamaludin, Ili Shairah Abdul Halim, Siti Lailatul Mohd Hassan, and Wan Fazlida Hanim Abdullah. Real-time FPGA-based sound source localization. In *2024 IEEE 14th Symposium on Computer Applications & Industrial Electronics (ISCAIE)*. IEEE, 2024. 1, 2

[17] Seunghun Jin; Dongkyun Kim; Hyung Soon Kim; Chang Hoon Lee; Jong Suk Choi; Jae Wook Jeon. Real-time sound source localization system based on FPGA. In *The IEEE International Conference on Industrial Informatics (INDIN 2008).*, 2008. 1, 2

[18] Y. Kadhim and A. Khalifeh. Performance evaluation and comparative analysis for widely used sound source localization algorithms. In *Proc. 2022 5th Int. Conf. Communications, Signal Processing, and their Applications (ICCSPA)*, 2022. 3

[19] U. Klein and T. Q. Võ. Direction-of-arrival estimation using a microphone array with the multichannel cross-correlation method. In *Proc. 2012 Int. Symp. Communications and Information Technologies*, pages 251–256, 2012. 4

[20] C. Knapp and G. Carter. The generalized correlation method for estimation of time delay. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 24(4):320–327, 1976. 1

[21] A. Kovalyov, K. Patel, and I. Panahi. Accurate real-time estimation of 2-dimensional direction of arrival using a 3-microphone array. In *Proc. 2023 57th Asilomar Conf. Signals, Systems, and Computers*, pages 1179–1183, 2023. 1, 2, 4

[22] Konrad Kowalczyk, Szymon Wozniak, Tomasz Chyrowicz, and Roman Rumian. Embedded system for acquisition and enhancement of audio signals. In *2016 Signal Processing: Algorithms, Architectures, Arrangements, and Applications (SPA)*, pages 68–71, Poznan, Poland, 2016. 2, 4

[23] Mauricio Kugler Teemu Tossavainen Susumu Kuroyanagi. Design of a compact sound localization device on a standalone FPGA-based platform. In *IEICE TRANS. INF. SYST*, 2016. 1, 2

[24] Krzysztof Kakol and Bożena Kostek. A study on signal processing methods applied to hearing aids. In *2016 Signal Processing: Algorithms, Architectures, Arrangements, and Applications (SPA)*, pages 219–224, 2016. 1

[25] Ákos Lédeczi, András Nádas, Péter Völgyesi, György Balogh, Branislav Kusy, János Sallai, Gábor Pap, Sebestyén Dóra, Károly Molnár, Miklós Maróti, and Gyula Simon. Countersniper system for urban warfare. *ACM Transactions on Sensor Networks*, 1(2):153–177, 2005. 2, 4

[26] J. A. Lopez-Olvera, H. M. Perez-Meana, E. Garcia-Rios, E. Escamilla-Hernandez, J. M. Carrichi-Chavez, and J. Betancourt-Martinez. Multichannel acoustic beamforming for speaker localization and DOA-based tracking. *Electronics*, 15(15):3410, 2026. 4

[27] Duy Nguyen, Parham Aarabi, and Ali Sheikholeslami. Real-time sound localization using field-programmable gate arrays. In *Proceedings of the 2003 IEEE International Conference on Multimedia and Expo (ICME)*, volume 2, pages 6–10, 2003. 1, 2

[28] U. T. K. Nguyen and T. V. Pham. Performance assessment of generalized cross-correlation based algorithms for multisource point-based localization and detection. In *Proc. 2011 Int. Conf. Advanced Technologies for Communications (ATC)*, pages 303–306, 2011. 3

[29] Lucian Petrica. An evaluation of low-power microphone array sound source localization for deforestation detection. *Applied Acoustics*, 113:162–169, 2016. 2, 4

[30] Lucian Petrica and Gheorghe Stefan. Energy-efficient wsn architecture for illegal deforestation detection. *International Journal of Sensors and Sensor Networks*, 3(3):24–30, 2015. 2, 4

[31] T. A. Powers and V. Hamacher. Three-microphone instrument is designed to extend benefits of directionality. *Hearing J.*, 55(10):38–45, Oct. 2002. 3

[32] H. Puder. Adaptive signal processing for interference cancellation in hearing aids. *Signal Process.*, 86(6):1239–1253, June 2006. 3

[33] K. Samtani, J. Thomas, S. P. Deepu, and S. David S. Area and power optimised ASIC implementation of adaptive beamformer for hearing aids. In *Proc. 2017 IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, pages 1–4, Turin, Italy, Oct. 2017. 3, 4

[34] K. Samtani, J. Thomas, G. A. Varma, S. David S., and S. P. Deepu. FPGA implementation of adaptive beamforming in hearing aids. In *Proc. 2017 39th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, pages 2239–2242, Seogwipo, South Korea, July 2017. 1, 2, 3, 4

[35] Tomyslav Sledevič and Raimond Laptik. An evaluation of hardware-software design for sound source localization based on soc. In *Proceedings of the 2017 Open Conference of Electrical, Electronic and Information Sciences (eStream)*, pages 1–4, Vilnius, Lithuania, 2017. 2, 4

[36] Tomoya Takagi, H. Noguchi, Koji Kugata, Masahiko Yoshimoto, and H. Kawaguchi. Microphone array network for ubiquitous sound acquisition. In *2010 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1474–1477, Dallas, TX, USA, 2010. 2, 4

[37] S. Tokgöz, A. Kovalyov, and I. Panahi. Real-time estimation of direction of arrival of speech source using three microphones. In *Proc. 2020 IEEE Workshop Signal Process. Syst. (SiPS)*, pages 1–5, 2020. 4

[38] S. Tokgoz and I. M. S. Panahi. Robust three-microphone speech source localization using randomized singular value decomposition. *IEEE Access*, 9:157800–157811, 2021. 4

[39] Trình Quc Võ and Udo Klein. A soft-processor-based FPGA implementation of a microphone array direction-of-arrival estimator. In *Vietnam National University HCMC.-unpublished*, 2013. 1, 2

[40] D. Ying, J. Li, Y. Feng, and Y. Yan. Direction of arrival estimation based on weighted minimum mean square error. In *Proc. 2013 IEEE China Summit and Int. Conf. Signal and Information Processing (ChinaSIP)*, pages 318–321, 2013. 3, 6